

Math 242, S13

Quiz 2

Name: Answer

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find a general solution for each of the following first-order differential equations.

(a) $\frac{dy}{dx} = \frac{1}{\sqrt{2x+1}}$.

$$y(x) = \int \frac{1}{\sqrt{2x+1}} dx$$
$$= \sqrt{2x+1} + C$$

(b) $\frac{dy}{dx} = 2x \sec(y)$.

$$\frac{dy}{dx} = 2x \sec(y)$$

$$\frac{1}{\sec y} dy = 2x dx$$

$$\cos y dy = 2x dx \Rightarrow \int \cos y dy = \int 2x dx$$

$$\sin y = x^2 + C$$

$$y = \arcsin(x^2 + C)$$